

# SANKALP ARUN KARKERA

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## PROFESSIONAL SUMMARY

Data engineer with 7+ years of experience designing and scaling data pipelines, now extending that foundation into product ownership for Risk Products within the Credit Risk & Fraud division. Expertise in large-scale data processing and driving insights for informed decision-making, with the ability to translate complex regulatory and business needs into actionable delivery plans.

## PROFESSIONAL EXPERIENCE

### Product Manager | *American Express*

July 2026 – Present

- Owned the product vision and strategy for Exposure product, which is a foundational capability in the Credit Risk & Fraud division that determines a customer's purchase capacity, shaping how authorization risk is assessed across the network in real time.
- Defined the roadmap for exposure calculation and balance management capabilities, prioritizing features that reduce false declines at the point of sale while protecting the business from unmanaged credit risk.
- Translated ambiguous business and regulatory needs into a structured feature roadmap, working with Engineering to scope and sequence capabilities that identify, calculate, and manage customer exposure balances.
- Participated in sprint planning, backlog grooming, and PI planning, coordinating dependencies across Engineering and Strategy teams on the Exposure roadmap.
- Defined success metrics for exposure management features to guide prioritization and measure the impact of new capabilities post-launch.

### Data Engineer | *Marriott International*

Nov 2019 – June 2026

- Designed and implemented scalable data architecture for Consumer Analytics domain that improved data processing efficiency by 40%, resulting in faster access to critical business insights.
- Designed, built, and managed pipelines using Python and PySpark to ingest, transform, and load sales, loyalty/rewards, and reservation data from multiple upstream systems into Snowflake, supporting downstream analytics and reporting.
- Architected and orchestrated Airflow DAGs to schedule and monitor batch workflows across sales, rewards, and reservations domains, handling dependencies, retries, and SLA-driven data requirements.
- Built real-time streaming pipelines using Kafka to capture and process reservation and rewards events, enabling near-real-time updates to downstream jobs and end-user analytics dashboards.
- Implemented data quality checks and validation across batch and streaming pipelines to catch upstream data issues before they reached reporting layers.
- Led architecture discussions for new ETL pipelines and provided technical guidance and informal mentorship to junior data engineers on PySpark best practices, SQL optimization, and Airflow DAG design.
- Monitored pipeline health and led root-cause analysis for production incidents affecting loyalty data feeds, improving pipeline reliability and reducing recurring failures.

### Software Engineer | *Infogain Corporation*

Jan 2016 – Apr 2017

- Developed and maintained Python-based backend services supporting the virtual data room (VDR) platform, handling document processing, access permissions, and workflow automation for enterprise clients.
- Debugged and resolved production issues related to document indexing, permission enforcement, and data integrity, improving platform reliability for high-stakes M&A and due diligence transactions.
- Used Git for version control across feature branches, participated in code reviews, and followed pull-request workflows to maintain code quality in a regulated, security-sensitive environment.
- Wrote unit and integration tests to validate business logic and prevent regressions in permission-handling and document-processing modules.

## EDUCATION

M.S. Information Systems — *New Jersey Institute of Technology*

2017 – 2019

B.E. Computer Science — *University of Mumbai*

2011 – 2015

## TECHNICAL SKILLS

**Languages:** Python, SQL, PL/SQL, PySpark, Shell scripting

**Cloud Services:** AWS (S3, EC2, Redshift, Lambda, Bedrock), GCP (BigQuery, Dataflow, Pub/Sub, Vertex AI)

**Databases:** Snowflake, BigQuery, PostgreSQL, DB2, SQL Server, MongoDB

**Big Data Technologies:** Hive, Kafka, Airflow, Snowflake, Apache NiFi, Tableau

**DevOps & Tooling:** Docker, Kubernetes, Jenkins, Git, JIRA, UNIX, Putty, ServiceNow

**AI Frameworks:** Semantic Modelling, LangChain, LangGraph, RAGs, Hugging Face Transformers